

Chapter 7 Secondary Storage

Introduction

-
- Data storage has expanded from text and numeric files to include digital music files, photographic files, video files, and much more.
 - These new types of files require secondary storage devices with much greater capacity.
 - In this chapter, you learn about the many types of secondary storage devices, including their capabilities and limitations.

Storage

-
- Primary storage is:
 - Volatile storage
 - Loses content when the computer loses power
 - Temporary storage
 - Random Access Memory (RAM)
 - Secondary storage is:
 - Nonvolatile storage
 - Stores programs and data regardless of power
 - Permanent storage
 - Permanently saves information for future use

Secondary Storage Characteristics

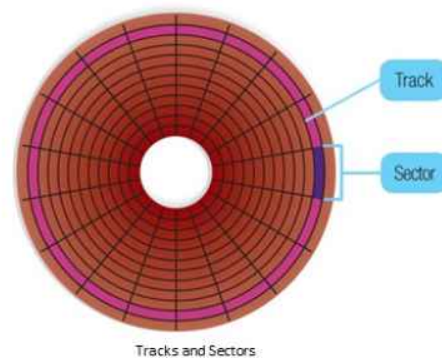
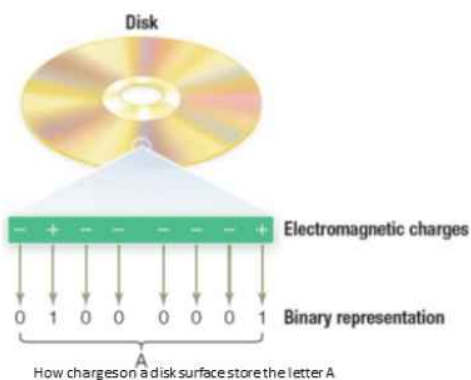
- Secondary storage characteristics
 - Media
 - Physical material that holds data and programs
 - Capacity
 - Measures how much the media can hold
 - Storage devices
 - Hardware that reads data and programs
 - Access speed
 - Amount of time required to retrieve data from storage
 - Writing is the process of saving information to storage
 - Reading is the process of accessing information from storage



Hard Disks

Save files by altering the magnetic charges of the disk's surface to represent 1s and 0s

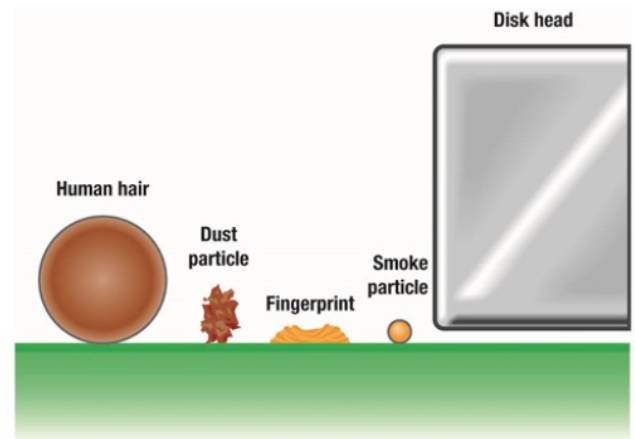
- Use rigid, metallic platters that are stacked one on top of one another
- Store and organize files using tracks, sectors, and cylinders



Head Crash

Occurs when read-write head makes contact with the hard disk's surface or with particles on its surface

- Disastrous



Types of Hard Disks

- Internal
 - Located inside the system unit
 - Used to store programs and data files
 - You should perform routine maintenance and periodically backup all important files
- External
 - Removable
 - Used to complement internal hard disk
 - Used to back up the contents of the internal hard disk



Performance Enhancements

There are 3 ways to enhance performance.

Technique	Description
Disk caching	Users cache and anticipates data needs
RAID	Linked, inexpensive hard-disk drives
File compression	Reduces file size
File decompression	Expands compressed files



Solid-State Storage

Solid-state storage devices (SSDs) have no moving parts

- Solid-state drives
 - Faster and more durable than hard disks
 - Access to slash memory or solid state storage
- Flash memory cards
 - Widely used in laptops, smartphones, GPS navigation systems
- USB Drives (or Flash Drives)
 - Connect to USB port
 - Capacity of 1 GB to 256 GB
 - Portable



Optical Discs

- Hold over 128 gigabytes (GB) of data
- Use reflected light to represent data
 - Lands represent 1s and 0s on the disc
 - Pits are bumpy areas on the disc that, when light is reflected, determine the 1s and 0s
- Use tracks and sectors to organize and store files but only use a single track unlike the hard drive

Optical Disc Formats and Types

Format	Total Capacity	Description
CD	700 MB	Once the standard optical disc
DVD	4.7 GB	Current standard
BD	50 GB	Hi-def format, large capacity

Type	Access	Description
ROM	Read only-memory	Cannot be written to or erased
R	Recordable	Can be written to
RW or RAM	Rewritable or random-access memory	Can be written to and erasable

Cloud Storage

The Internet acts as a “cloud” of servers

- Applications provided as a service rather than a product
- Supplied by servers that provide cloud storage or online storage

Cloud Storage Services

Advantages

- Maintenance
- Hardware upgrades
- File sharing and collaboration

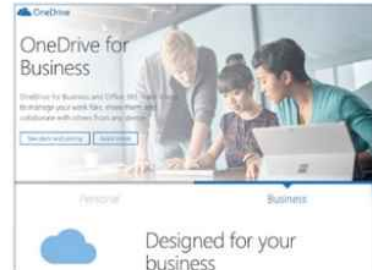
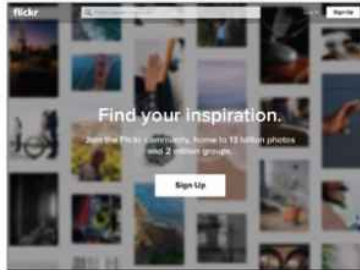
Disadvantages

- Access speed
- File security

Company	Location
Dropbox	www.dropbox.com
Google	Drive.google.com
Microsoft	onedrive.live.com
Amazon	Amazon.com/cloud
Apple	www.icloud.com

Making IT Work for You ~ Cloud Storage

Using a cloud storage service makes it easy to upload and share files with anyone.



Mass Storage Devices

To meet the needs of organizations requiring large amounts of secondary storage requirements

- Enterprise storage system
 - Safe use of data across an organizational network
- Devices include:
 - File servers
 - Networked attached storage (NAS)
 - RAID systems
 - Organizational cloud storage



Storage Area Network (SAN)

- Architecture to link remote computer storage devices
- User's computer provides file system, but SAN provides disk space
- House data in remote locations and still allow efficient and secure access

Careers In IT

- Disaster recovery specialists are responsible for recovering systems and data after a disaster strokes
- General employer requirements
 - Bachelor's or associate's degree in computer science or information systems
 - Experience in the field and skills in networking, security and DBA
 - Communication and skills and be able to handle high-stress situations
- Annual salary of \$73 K to \$92 K



A Look to the Future ~ Next Generation Storage

- In the future, your photos and videos may be stored using heat, lasers, or even DNA
- Both magnetic disk manufacturers and flash memory researchers are working on increasing storage without increasing physical size